

1. Decomposition

A large task can be divided by phase, by subsystem, by uncertainty, or by evidence source. Only some decompositions support parallel agents. The key question is whether two pieces can make progress without needing the same mutable state or the other piece's result.

Good parallel candidates include independent repository exploration, comparison of alternative implementations, review of separate modules, or research into distinct sources. Poor candidates include two agents editing the same function, tasks whose requirements are still changing, or a chain in which each step depends on the previous output.

2. Why Subagents

OpenAI's Codex documentation supports specialized subagents that can work in parallel and return results to the parent. Astra is trained to divide and delegate work when the harness makes that available. The potential gains are elapsed time, independent lines of inquiry, and reduced context pollution in the root thread.

The costs are duplicated context, additional tokens, coordination overhead, merge risk, and the possibility that local success does not compose into global success.

3. The Independence Test

Before delegating A and B in parallel, ask:

- State: Do they write the same files or depend on the same mutable external state?
- Inputs: Can each start from the current evidence without waiting for the other?
- Contract: Can each return a result with an explicit interface or conclusion?
- Merge: Can the parent combine them without recreating most of the work?
- Verification: Can each sub-result be checked independently?

If several answers are no, sequential work is probably simpler.

4. Role Specialization

Subagents can also separate objectives. One agent can explore without editing; another can implement after receiving a map; another can review the diff. This is useful when mixing the objectives would create premature commitment. An explorer asked both to find and fix may stop searching as soon as it finds one plausible patch location.

Specialization should remain bounded. An "expert" role that merely adds flattering adjectives is not a capability. Give the subagent a distinct evidence source, scope, output contract, or tool permission.

5. Delegation Prompt

A useful delegated task contains:

Object: exactly what part of the system to inspect or change.

Question: what the parent needs answered.

Constraints: files, interfaces, or actions that must not change.

Evidence: commands, references, or observations to use.

Return shape: findings, patch, test result, or recommendation.

Avoid asking three agents "solve the whole problem independently" unless comparison is the purpose. That multiplies cost and often produces three large, incompatible answers.

6. Parent Responsibility

Delegation does not transfer accountability. The root agent must compare returned claims against the original task, detect conflicts, decide which evidence is strongest, and verify the integrated result. A subagent saying "tests pass" is not the same as the parent knowing which tests ran against which state.

7. When One Agent Wins

One Astra thread is often better when the task is small, tightly coupled, or requires a single evolving mental model. Additional agents can make simple work slower by turning local reasoning into a coordination problem. Parallelism is a means, not a mark of sophistication.

References: OpenAI, Codex subagents; OpenAI, GPT-6 Astra model guidance; OpenAI, Codex.

8. Worked Decomposition

Task: replace a legacy authentication library across four services and preserve login behavior. A weak decomposition assigns one service to each agent immediately. That assumes the migration contract is already known. A stronger first phase delegates independent discovery: one agent maps library usage, one compares new-library semantics, and one inventories authentication tests. The parent then defines the shared migration contract. Only after that should service-level edits proceed in parallel. Parallelism is delayed until the interface is stable.

9. Competing Hypotheses

Subagents can be used for intellectual diversity rather than file parallelism. Give Agent A the hypothesis that the bug is a race, Agent B the hypothesis that stale cache state is responsible, and Agent C a read-only trace of the request path. Require each to return the observation that would falsify its hypothesis. The parent can then choose the cheapest discriminating experiment. This is more useful than asking three agents for three complete fixes.

10. Coordination Failure

An agent can return a locally correct result that is unusable because the return contract was vague. "I investigated the parser" forces the parent to reopen the whole branch. "The failure begins in normalizeToken at parser.ts:118; input X produces Y before the patch; test command Z reproduces it" gives the parent a reusable piece of evidence. Delegation quality depends on the interface between agents as much as on intelligence inside each agent.

Independent Open Course

<https://github.com/dicnunz/codex-astra-course>

CODEX.00 | Codex with GPT-6 Astra

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